```
> restart;#Audion, the phonon Planck's law;
> ec := 1.602176634*10^(-19);
  va := 343;
  c := 299792458;
```

$$ec := 1.602176634 \times 10^{-19}$$

$$va := 343$$

$$c := 299792458 \quad (1)$$

```
> E1 := 0.00762014687*t + a*t^2 + b*t^3 + d*t^4 + f*t^5;
```

$$E1 := 0.00762014687\,t + a\,t^2 + b\,t^3 + d\,t^4 + f\,t^5 \quad (2)$$

```
> E2 := subs(t = 435, E1);
```

$$E2 := 3.314763888 + 189225\,a + 82312875\,b + 35806100625\,d + 15575653771875\,f \quad (3)$$

```
> E3 := subs(t = 479, E1);
```

$$E3 := 3.650050351 + 229441\,a + 109902239\,b + 52643172481\,d + 25216079618399\,f \quad (4)$$

```
> E4 := subs(t = 528, E1);
```

$$E4 := 4.023437547 + 278784\,a + 147197952\,b + 77720518656\,d + 41036433850368\,f \quad (5)$$

```
> E5 := subs(t = 2860, E1);
```

$$E5 := 21.79362005 + 8179600\,a + 23393656000\,b + 66905856160000\,d + 191350748617600000\,f \quad (6)$$

```
> a := -1.44670357887361*10^(-7);b := -1.049267156*10^(-9); d :=
  1.890440485*10^(-12);f :=-6.233924848*10^(-16);
```

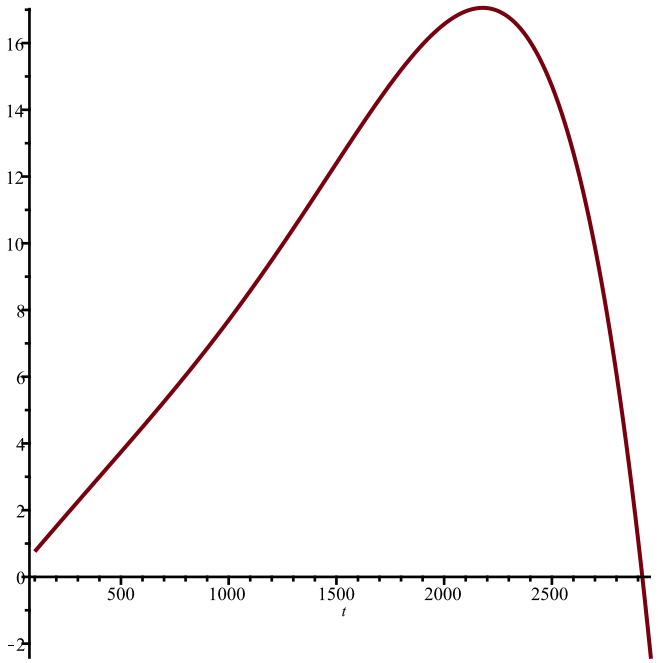
$$a := -1.446703579 \times 10^{-7}$$

$$b := -1.049267156 \times 10^{-9}$$

$$d := 1.890440485 \times 10^{-12}$$

$$f := -6.233924848 \times 10^{-16} \quad (7)$$

```
> plot(E1,t=100..2960);solve(3.259 = E1, t);solve(3.59 = E1, t);solve
  (3.95 = E1, t);E1;
```



434.9999999, 2860.000000, 579.9964863 + 1617.918467 I, −1422.488521, 579.9964863 − 1617.918467 I

479.6275541, 2853.405326, 565.9474392 + 1617.810863 I, −1432.423307, 565.9474392 − 1617.810863 I

528.0000000, 2846.108503, 550.6640041 + 1618.311176 I, −1442.932060, 550.6640041 − 1618.311176 I

$$0.00762014687\,t - 1.446703579\times10^{-7}\,t^2 - 1.049267156\times10^{-9}\,t^3 + 1.890440485\times10^{-12}\,t^4 - 6.233924848\times10^{-16}\,t^5 \quad (8)$$

```
> restart;#The Phonon Planck's "constant"
> restart;#Planck's law for Phonon;
> with(PDEtools, dchange):with(plots):
> with(Units):
```

```
Automatically loading the Units[Simple] subpackage
```

```
> kernelopts(maxdigits);
```

$$38654705646 \quad (9)$$

```
> Digits:=10;
```

$$Digits := 10 \quad (10)$$

```
> params:={k= 1.380649*10^(-23)*Unit(J/K),rb=5.293*10^(-11)*Unit(m),he=
  4.135667697*10^(-15)*Unit(eV*s),
  ec= 1.602176634*10^(-19)*Unit(C),Tq=1.765*10^(-19)*Unit(s),
  c= 299792458*Unit(m/s),T=297*Unit(K),nu1=7.880979442*10^14*Unit(s^(-1)
  ),E1=3.259*Unit(eV)};
```

$$params := \left\{E1 = 5.221493608\times10^{-19}\ \text{J},\ T = 297\ \text{K},\ Tq = 1.765000000\times10^{-19}\ \text{s},\ c = 299792458\ \frac{\text{m}}{\text{s}},\ ec = 1.602176634\times10^{-19}\ \text{C},\ he = 6.626070096\times10^{-34}\ \frac{\text{m}^2\,\text{kg}}{\text{s}},\ k = 1.380649000\times10^{-23}\ \frac{\text{m}^2\,\text{kg}}{\text{s}^2\,\text{K}},\ vl = 7.880979442\times10^{14}\ \frac{1}{\text{s}},\ rb = 5.293000000\times10^{-11}\ \text{m}\right\} \quad (11)$$

```
> NV:=(c/va)*32/3*Pi*Tq1*nu^2*ha*nu/(exp(ha*nu/(k*T) - 1));
```

$$NV := \frac{32\,c\,\pi\,Tq1\,\nu^3\,ha}{3\,va\,e^{\frac{ha\,\nu}{kT}} - 1} \quad (12)$$

```
> E:=diff(NV,nu);
```

$$E := \frac{32\,c\,\pi\,Tq1\,\nu^2\,ha}{va\,e^{\frac{ha\,\nu}{kT}} - 1} - \frac{32\,c\,\pi\,Tq1\,\nu^3\,ha^2}{3\,va\,e^{\frac{ha\,\nu}{kT}} - 1\,kT} \quad (13)$$

```
> solve(E=0);
```

$$\{T = T, Tq1 = Tq1, c = c, ha = ha, k = k, \nu = 0, va = va\},\ \{T = T, Tq1 = Tq1, c = c, ha = 0, k = k, \nu = \nu, va = va\},\ \{T = T, Tq1 = Tq1, c = 0, ha = ha, k = k, \nu = \nu, va = va\},\ \{T = T, Tq1 = 0, c = c, ha = ha, k = k, \nu = \nu, va = va\},\ \left\{T = \frac{ha\,\nu}{3\,k}, Tq1 = Tq1, c = c, ha = ha, k = k, \nu = \nu, va = va\right\} \quad (14)$$

```
> #Thus, for k=3/2*k*T we have the solution
> c:=299792458;nu1:=435;c/nu1;ec:= 1.602176634*10^(-19);k:= 1/2;E1:=
  3.259*ec;T:=297;va:=343;Tq1:=c/va*1.765*10^(-19);ha:=0.007486790719;
  ha*ec;
```

$$c := 299792458$$

$$vl := 435$$

$$\frac{299792458}{435}$$

$$ec := 1.602176634\times10^{-19}$$

$$k := \frac{1}{2}$$

$$E1 := 5.221493650 \times 10^{-19}$$

$$T := 297$$

$$va := 343$$

$$Tq1 := 1.542663814 \times 10^{-13}$$

$$ha := 0.007486790719$$

$$1.199516115 \times 10^{-21} \quad \textbf{(15)}$$

```
> NV:=c/va*32/3*Pi*Tq1*nu^2*ha*nu/(exp(ha*nu/(k*T) - 1));
```

$$NV := \frac{3.382767620 \times 10^{-8}\, \nu^3}{e^{0.00005041609912\,\nu} - 1} \quad \textbf{(16)}$$

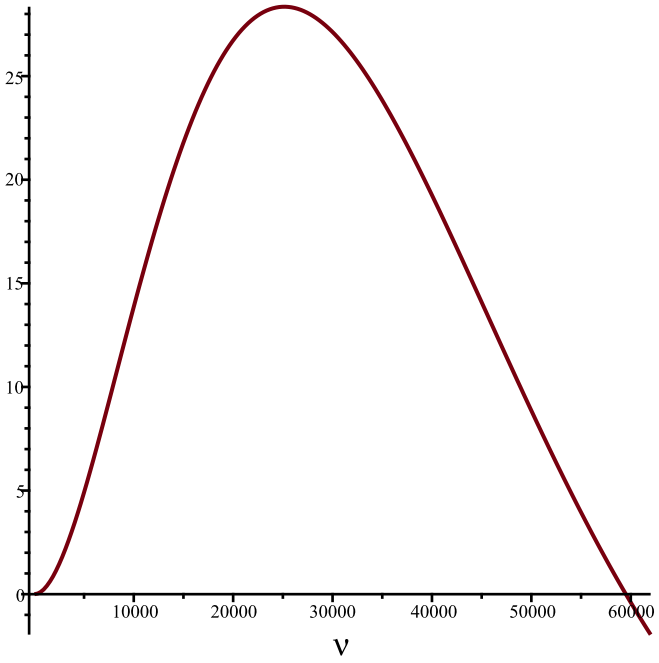
```
> E:=diff(NV,nu);
```

$$E := \frac{1.014830286 \times 10^{-7}\, \nu^2}{e^{0.00005041609912\,\nu} - 1} - \frac{1.705459476 \times 10^{-12}\, \nu^3}{e^{0.00005041609912\,\nu} - 1} \quad \textbf{(17)}$$

```
> solve(E=0);
```

$$0., 59504.80209 \quad \textbf{(18)}$$

```
> plot(E,nu=10..6.2*10^4);
```




```
>
```